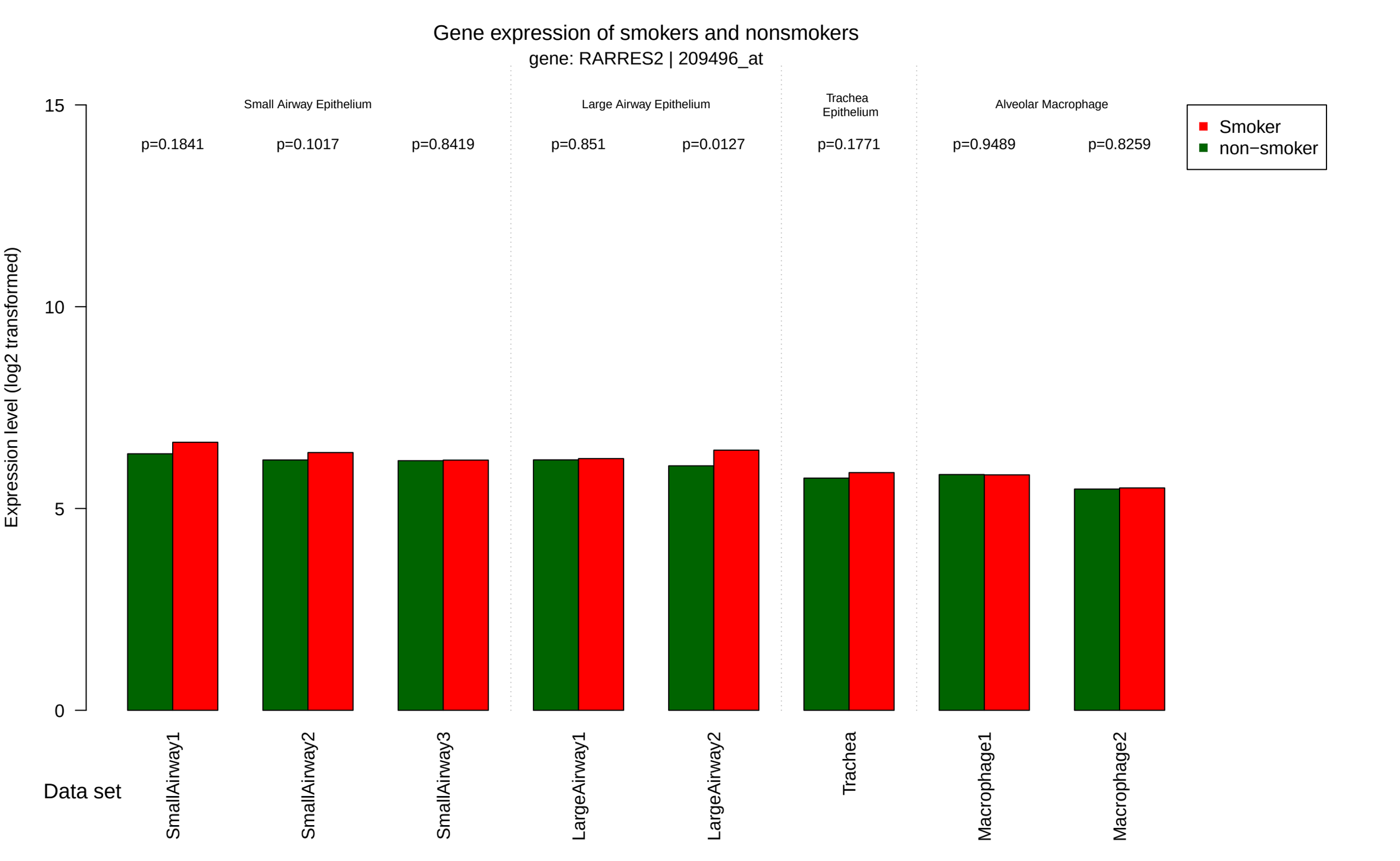
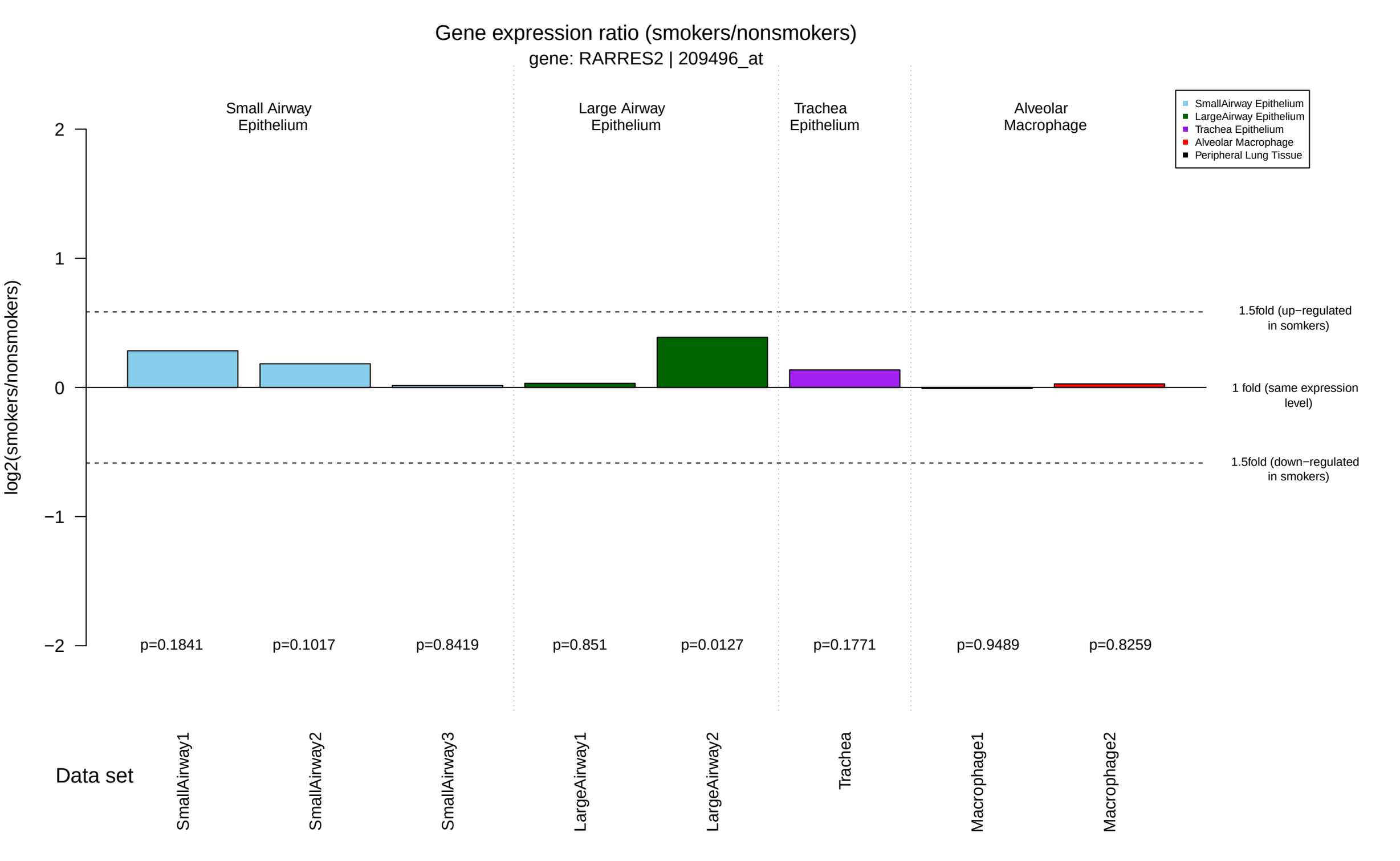
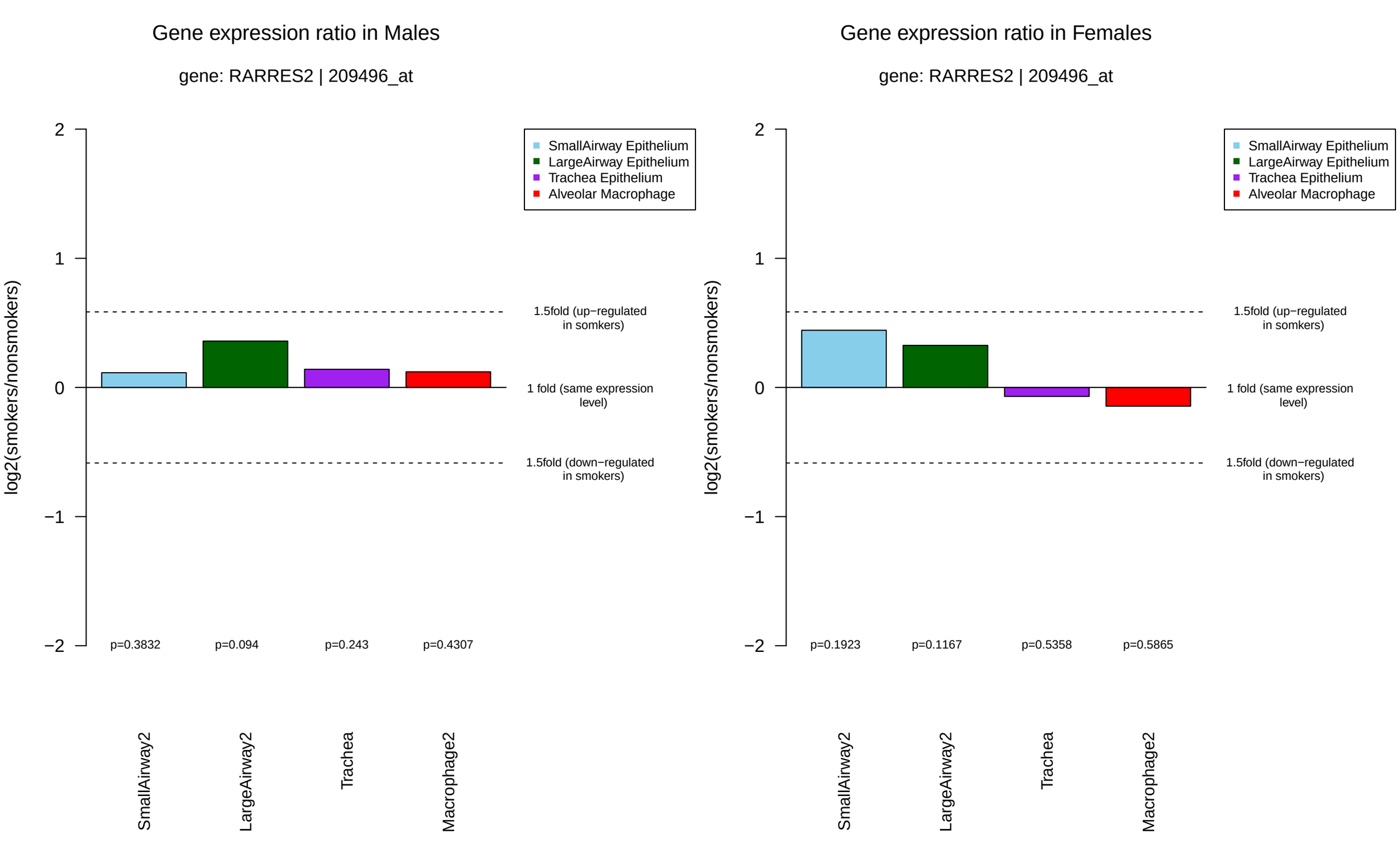
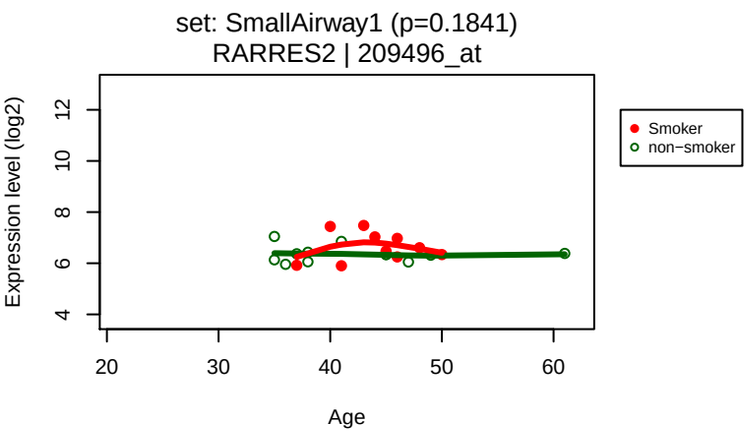
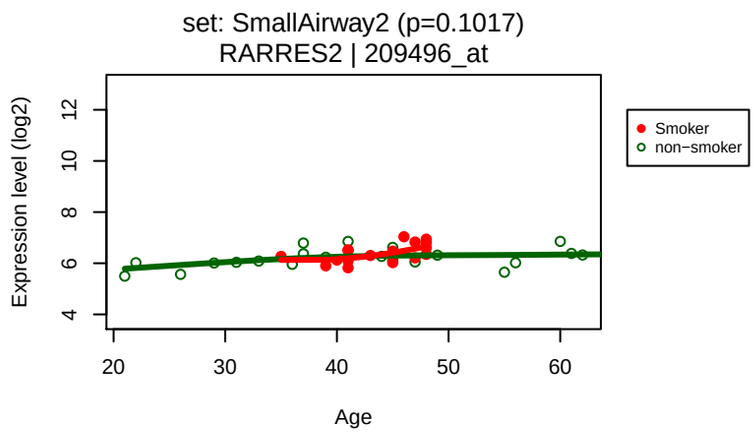
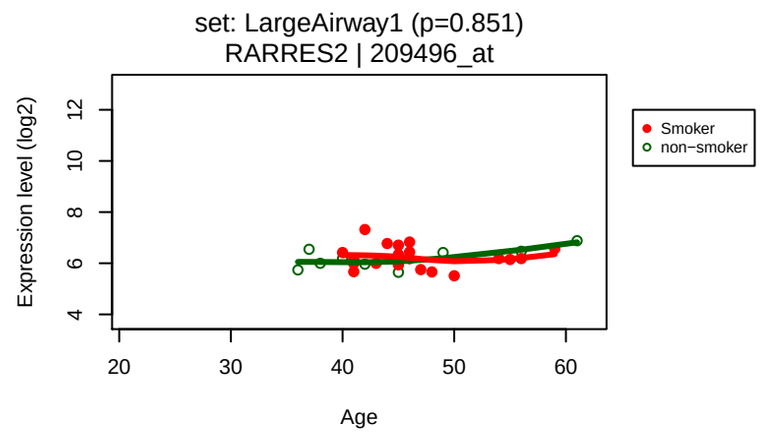

Gene expression of smokers and nonsmokers
gene: RARRES2 | 209496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1841
p=0.1017
p=0.8419
p=0.851
p=0.0127
p=0.1771
p=0.9489
p=0.8259
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

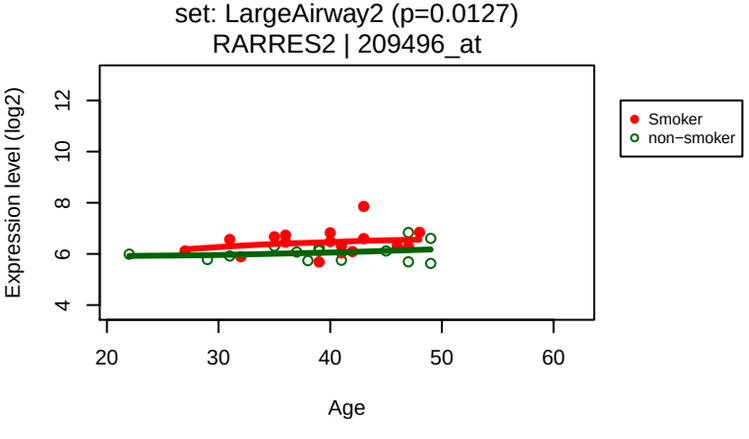

Gene expression ratio (smokers/nonsmokers)
gene: RARRES2 | 209496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1841
p=0.1017
p=0.8419
p=0.851
p=0.0127
p=0.1771
p=0.9489
p=0.8259
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

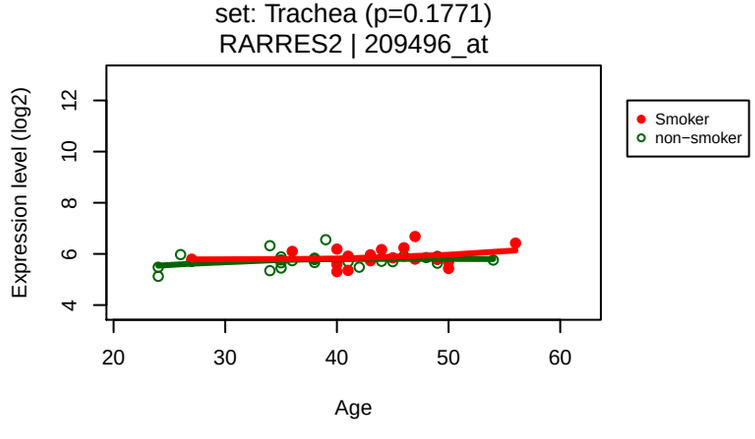

Gene expression ratio in Males
gene: RARRES2 | 209496_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3832
p=0.094
p=0.243
p=0.4307
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARRES2 | 209496_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1923
p=0.1167
p=0.5358
p=0.5865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

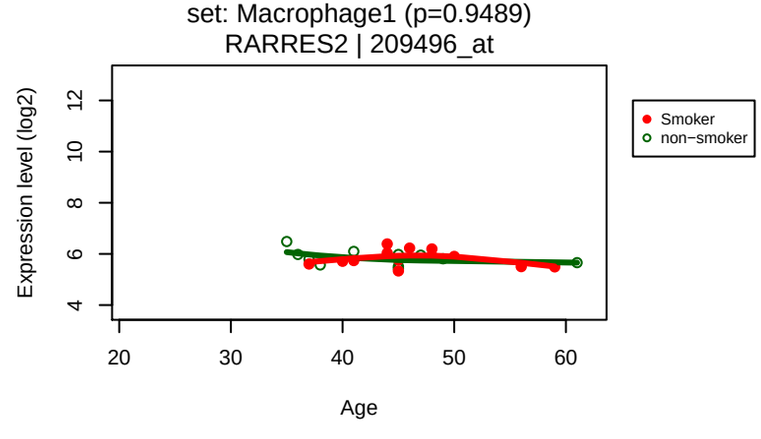

set: SmallAirway1 (p=0.1841)
RARRES2 | 209496_at
Expression level (log2)
Age
Smoker
non−smoker

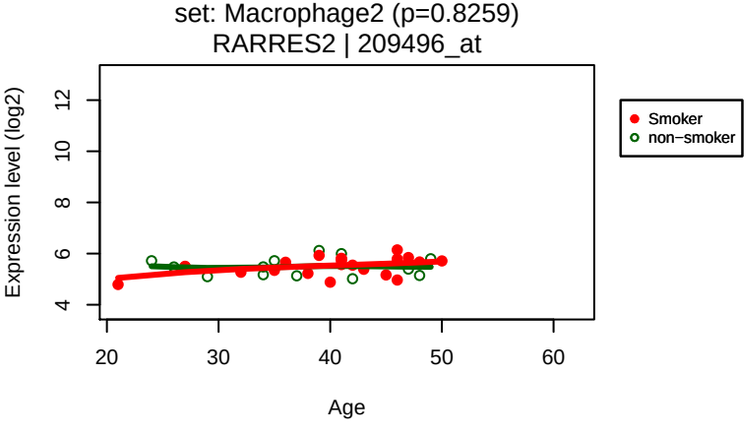

set: SmallAirway2 (p=0.1017)
RARRES2 | 209496_at
Expression level (log2)
Age
Smoker
non−smoker

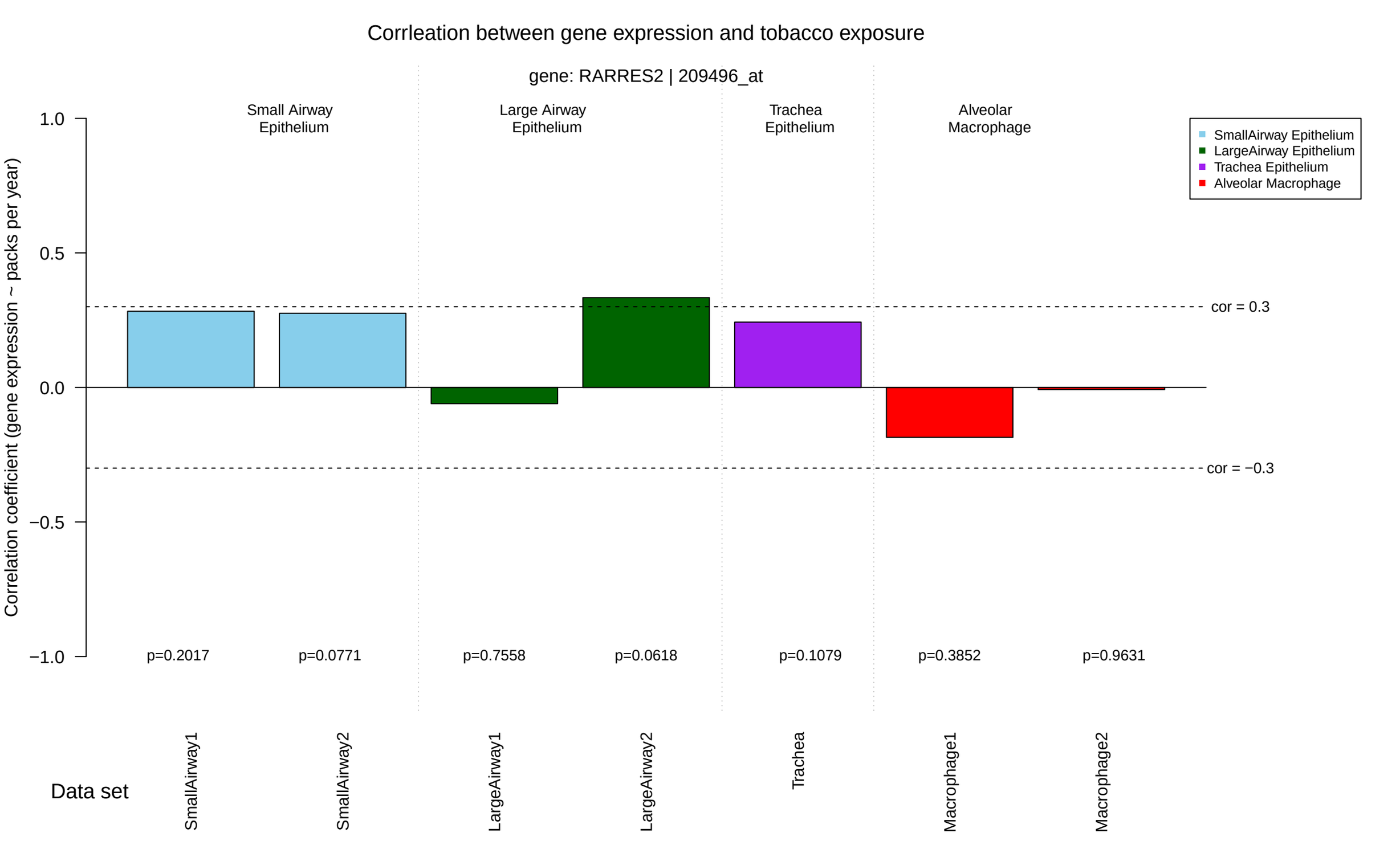

set: LargeAirway1 (p=0.851)
RARRES2 | 209496_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0127)
RARRES2 | 209496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1771)
RARRES2 | 209496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9489)
RARRES2 | 209496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8259)
RARRES2 | 209496_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARRES2 | 209496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2017
p=0.0771
p=0.7558
p=0.0618
p=0.1079
p=0.3852
p=0.9631
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2